Aniket Pratap

858-353-6626 | aniketpx@gmail.com | https://aniketpratap.netlify.app/

Education

University of California San Diego
Master of Computer Science — *Expected Dec 2025*

University of California Irvine — 3.83 GPA
Bachelor of Computer Science — *Graduated Jun 2024*

Experience

Algorithms Intern — June 2025 - Sept 2025
KLA - MACH Team — *Ann Arbor, MI*

- Achieved a 100× speedup by converting a sequential C++ wafer-etching simulation into a parallel CUDA GPU pipeline, reducing runtimes from hours to minutes
- Architected the entire GPU implementation (kernel design, memory tiling, race-condition prevention), enabling the team to run high-resolution physics simulations previously infeasible on CPU
- Built CPU vs. GPU performance heatmaps adopted for ongoing benchmarking, helping the team identify bottlenecks and select future optimization targets
- Refactored a legacy C++ simulation codebase to enable GPU interoperability and improve modularity, maintainability, and debugging speed, allowing faster iteration on physics models
- Coordinated with engineers across Milpitas, Ann Arbor, and Austin to reverse-engineer the simulation pipeline and ensure correctness across CPU/GPU implementations

Software Engineering Intern — Apr 2024 - Sept 2024
Ekastar — *Remote*

- Implemented a secure authentication workflow integrating Firebase and an Azure backend using token-based validation, improving login reliability and user-data consistency
- Improved web load speed by 70% via async fetching and optimized rendering, significantly enhancing user experience and reducing session drop-offs
- Worked in weekly product cycles with the CEO and backend team, rapidly iterating on UX flows and shipping customer-facing features in a fast-paced startup environment

NLP Researcher — June 2023 - Apr 2024
University of California Irvine — *Irvine, CA*

- Scraped and cleaned live smoke-cessation forum data and built an NLP pipeline to analyze user intention; deployed a real-time chatbot via AWS Lambda integrated with the GroupMe API
- Fine-tuned GPT-2 to reach 98% sentiment-classification accuracy and improved training speed by 40% through NumPy vectorization and optimized data pipelines

Projects

ZotClass + ZotAPI — Full-Stack Mobile & Backend Platform | https://github.com/xXViridianXx/ZotClass

- Built a React Native mobile app (Android/iOS) used by 1,000+ students to browse courses, plan schedules, and sync selections to their calendar
- Developed ZotAPI using Python web scraping and deployed distributed scrapers on AWS Lambda, improving data freshness and scraping speed by 99%
- Integrated the backend with the mobile app to deliver real-time class availability and prerequisite data backed by DynamoDB for low-latency queries
- Implemented fast, responsive in-app search and UI flows using TypeScript and React Native components, enabling seamless class-planning experiences across Android and iOS

Technical Skills

Languages: Python, C++, C#, JavaScript, SQL
Systems / Performance: CUDA, multithreading, parallel computing
Frameworks: React, React Native, Node.js, Flask, Django
ML/AI: PyTorch, TensorFlow, NumPy, Pandas, Matplot
Cloud/Tools: AWS (Lambda, DynamoDB), Firebase, Docker, Git,